

Makeup examination module 4

Section A Multiple Choice Questions (Total 40 marks)

1. $\iint_{D: x^2+y^2 \leq 1} \left(\frac{x^3}{1+y^2} - \frac{y}{x^2+2} + 1 \right) dx dy = (\quad)$

- (a) 0, (b) 1; (c) -2; (d) π

2. Suppose the function $z = f(ax + by, cx + dy)$, a, b, c, d are constants, f is of continuous

second order partial derivatives, then $\frac{\partial^2 z}{\partial x \partial y}$ can be expressed by

(a) $\frac{\partial^2 z}{\partial x \partial y} = abf_{11} + (ac + bd)f_{12} + cd \cdot f_{22}$

(b) $\frac{\partial^2 z}{\partial x \partial y} = abf_{11} + (ad + bc)f_{12} + cd \cdot f_{22}$

(c) $\frac{\partial^2 z}{\partial x \partial y} = a^2 f_{11} + (ad + bc)f_{12} + d^2 \cdot f_{22}$

(d) $\frac{\partial^2 z}{\partial x \partial y} = abf_{11} + (ad - bc)f_{12} + cd \cdot f_{22}$

3. Given a surface Σ with the equation $x^2 - 9y^2 + \frac{z^2}{4} = 1$, the tangent plane of the surface

Σ at point $P(3, 1, -2)$ can be expressed by the following equation

(a) $6x - 18y - z + 2 = 0$

(b) $3x - 9y - \frac{1}{2}z = 1$

(c) $6x - 18y + z + 2 = 0$

(d) $6x - 8y + z + 2 = 0$

4. The directional derivative of the function $f(x, y) = x^2 y^3 - y^5$, at point $P(1, 1)$ in the

direction $\vec{v} = (1, 1)$ is given by

(a) $\frac{\partial f}{\partial \vec{v}} = \frac{1}{2}$

(b) $\frac{\partial \vec{u}}{\partial \vec{v}}$

(c) $\frac{\partial \vec{u}}{\partial \vec{v}}$

(d) $\frac{\partial \vec{u}}{\partial \vec{v}}$

5. $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\sqrt{1-x^2-y^2}-1}{x^2+y^2} = (\quad)$

(a) 1, (b) 2; (c) 3, (d) $\frac{1}{2}$

6. $\iint_D \sqrt{R^2-x^2-y^2} d\sigma = (\quad)$ D is the disk with center the origin and radius R .

(a) π ; (b) $\frac{2}{3}\pi$; (c) $\frac{2}{3}\pi R^2$; (d) $\frac{2}{3}\pi R^3$

7. Given $z = x^3 \sin y$, then $\frac{\partial^2 z}{\partial x \partial y} = (\quad)$

(a) $\frac{\partial^2 z}{\partial x \partial y} = 3x^2 \cos y$

(b) $\frac{\partial^2 z}{\partial x \partial y} = -3x^2 \cos y$

(c) $\frac{\partial^2 z}{\partial x \partial y} = -\frac{x^4}{4} \cos y$

(d) $\frac{\partial^2 z}{\partial x \partial y} = \frac{x^4}{4} \cos y$

8. The tangent line about the curve $x=t, y=t^2, z=t^3$ at the point $P(1,1,1)$ is with the equation as follows.

(a) $(x-1) + 2(y-1) + 3(z-1) = 0$

(b) $\frac{(x-1)}{1} = \frac{(y-1)}{2} = \frac{(z-1)}{3}$

(c) $x + 2y + 3(z-1) = 0$

(d) $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$

9. In order to find the extreme values of the function $f(x, y) = x^2 + 2y^2$ on the circle

$x^2 + y^2 = 1$ by using the method of Lagrange Multiplier, the auxiliary function

$L(x, y, z, \lambda)$, can be given by

(a) $L(x, y, \lambda) = x^2 + 2y^2 + \lambda(1 + x^2 - y^2)$

(b) $L(x, y, \lambda) = \lambda(x^2 + 2y^2) + \lambda(1 - x^2 - y^2)$

(c) $L(x, y, \lambda) = x^2 + 2y^2 + \lambda(1 - x^2 - y^2)$

(d) $L(x, y, \lambda) = x^2 + 2y^2 + \lambda(1 + x^2 + y^2)$

10. The function $z = x^2 - 3y^2$ at the point $P(0, 0)$ achieves () .

(a) a local maximum value;

(b) a local minimum value;

(c) no maximum or minimum value;

(d) not sure

Section B Written Questions (Total 50 marks)

11. $\iiint_{\Omega} (x^2 + y^2) dV$, where Ω is the region bounded by the surface with equation

$x^2 + y^2 = 2z$ and a plane with equation $z = 2$.

12. Use geometry or symmetry or both , to evaluate the double integral

$$\iint_D (x+2) d\sigma, D = \{(x, y) \mid 0 \leq y \leq \sqrt{9-x^2}\}$$

13. Evaluating the integral by reversing the order of the integration

$$\int_0^1 dx \int_0^{\sqrt{x}} e^{-\frac{y^2}{2}} dy$$

14. Calculate the iterated integrals by using polar coordinates

$$\int_{-2}^2 dx \int_0^{\sqrt{4-x^2}} \sin(x^2 + y^2) dy$$

15. Draw the domain of the integration and then find the value of the following

$$\int_0^{\frac{a}{\sqrt{2}}} dx \int_0^x \sqrt{x^2 + y^2} dy + \int_{\frac{a}{\sqrt{2}}}^a dx \int_0^{\sqrt{a^2-x^2}} \sqrt{x^2 + y^2} dy$$

Section C (10 marks)

Suppose that $f(x) > 0$, prove that $\iint_{\substack{a \leq x \leq b \\ a \leq y \leq b}} \left[f(x) + \frac{1}{f(y)} \right] dx dy \geq 2(b-a)^2$